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Studies on *t*-BuOK-catalyzed Michael addition of 1,2-allenic ketones with 2-substituted diethyl malonates: highly selective synthesis of β,γ-unsaturated enones

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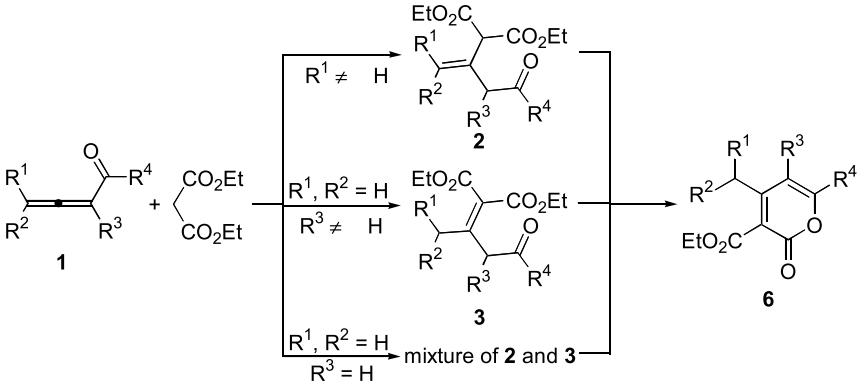
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Abstract—In this paper, we have demonstrated a facile nucleophilic addition of 2-substituted diethyl malonate to various substituted 1,2-allenic ketones to afford β,γ-unsaturated enones using a catalytic amount of *t*-BuOK as the base. The reaction usually completes within 10 min in acetone at room temperature. The stereoselectivity of the reaction with γ-substituted allenic ketones is very high affording the β,γ-unsaturated *E*-enones.
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1. Introduction

The Michael addition of 1,3-dicarbonyl compounds to α,β-unsaturated carbonyl compounds is an important reaction in organic synthesis,[1] as the product obtained are valuable building blocks in organic chemistry.[2] 1,2-Allenic ketones can be viewed as methylene-substituted α,β-unsaturated carbonyl compounds which constituted of two mutually vertical π-systems (π^2–π^4) rather than a plannar conjugated π-system + (π^6).[3] The attractive feature of these compounds confers on them an otherwise unattainable chemical reactivity and regioselectivity in organic synthesis.[4] β,γ-unsaturated enones can be found in some natural products[5] and are important intermediate in organic synthesis. Instead of β,γ-unsaturated enones, the Michael reaction to 1,2-allenyl ketones usually gives α,β-unsaturated enones.[6] In our previous work, we have demonstrated that diethyl malonate with various substituted 1,2-allenic ketones afforded β,γ-unsaturated enones **2** or **3** or their mixtures depending on where the substituents located (Scheme 1).[7] In our continuous effort for the synthesis of β,γ-unsaturated enones,[8] we propose that we may

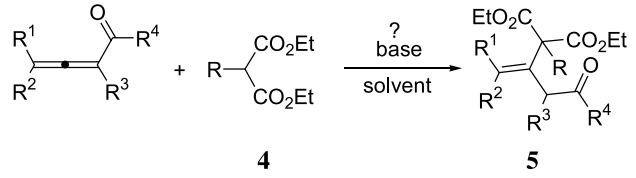


Scheme 1.

Keywords: Michael addition; 1,2-Allenic ketones; β,γ-Unsaturated enone; Catalysis.
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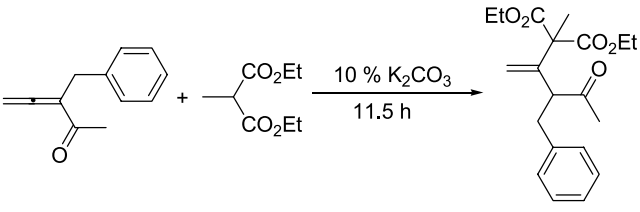
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exclusively get the β,γ-unsaturated enones **5** if a substitutent was introduced at the 2-position of diethyl malonate, which blocks the migration of carbon–carbon double bond to form **3**-type product (Scheme 2). However, the reaction of 3-benzyl-3,4-penten-2-one **1a** with 2-methyl malonate under the standard reaction condition we used in Ref. 7 was slow (Scheme 3). In this paper, we would like to disclose a general and highly efficient method for the synthesis of β,γ-unsaturated enones from the reaction of 1,2-allenic ketones and 2-substituted diethyl malonate.



4 **5**

Scheme 2.



1a **4a** **5aa**

solvent	Temp.	yield of **5aa**(%)
acetone	reflux	44[a]
DMF	80°C	27
EtOH	80°C	9

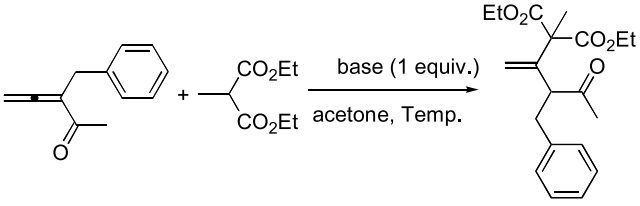
[a] 30% of **1a** was recovered

Scheme 3.

The Michael addition of 2-methyl diethyl malonate **4a** to 1,2-allenic ketone **1a** in different solvent under various base was investigated. The effect of solvents for the reaction was briefly investigated. Among the solvents tested (acetone, DMF, and EtOH), acetone gave the best yield (44%) (Scheme 3). Further studies showed that the yield was improved substantially when the reaction was carried out with 1 equiv of K_2CO_3 affording **5aa** in 75% yield (Table 1, entry 1). The base also affected the reaction greatly. For organic bases, such as NEt_3, pyridine and DABCO, no products were formed (Table 1, entries 2–4); the product **5aa** was obtained in low yields when NaOAc or NaOEt was used as the base (Table 1, entries 5–6); when strong base such as NaOH was used, **5aa** was formed in a moderate yield (Table 1, entry 7); However, the Cs_2CO_3-promoted Michael addition of **4a** to **1a** proceeded to yield **5aa** in 78% yield (Table 1, entry 8); the reaction also proceeded well in the presence of DBU producing **5aa** in the yield of 80% at room temperature in 10 min (Table 1, entry 9); It is interesting to find that with 10 mol% of *t*-BuOK the Michael addition of **1a** and **4a** was complete within 10 min to afford the product **5aa** in 82% yield (Table 1, entry 10). Thus, in following cases the reaction was conducted at room temperature in acetone using 10% *t*-BuOK as the catalyst.

The potential and scope of the *t*-BuOK (10 mol%)-catalyzed Michael addition is demonstrated by the reaction of 1,2-allenic ketones **1a–g** with 2-substituted diethyl malonate **4a–c**. The results are presented in Table 2. In a similar way, 2-allyl diethyl malonate **4b** and 2-benzyl diethyl malonate **4c** reacts with **1a** in the presence of 10% *t*-BuOK to afford the β,γ-unsaturated enones **5ab** and **5ac** in 85 and 81% yield, respectively (Table 2, entries 1 and 2). The Michael addition also proceeded smoothly with α-allyl-1,2-allenic ketone **1b** to give products **5ba**, **5bb** and **5bc** in 83, 78 and 76% yield, respectively (Table 2, entries 3–5). The reaction of **1c** with **4a** in the presence of 100 mol% *t*-BuOK in acetone instead of the standard 10 mol% *t*-BuOK

Table 1. Base-catalyzed michael addition of 2-benzyl-1,2-propadienyl methyl ketone **1a** with **4a**

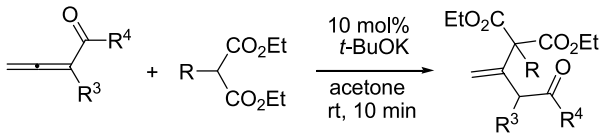


1a **4a** **5aa**

Entry	Base	Time	Temperature	Yield of **5aa** (%)
1	K_2CO_3	11.5 h	Reflux	75
2	NEt_3	12 h	Reflux	NR
3	Pyridine	12 h	Reflux	NR
4	DABCO	12 h	Reflux	NR
5	NaOAc	12.5 h	Reflux	5[a]
6	NaOEt	12.5 h	Reflux	4[b]
7	NaOH	12.5 h	Reflux	52
8	Cs_2CO_3	1 h	Reflux	78
9	DBU	10 min	rt	80
10	10% *t*-BuOK	10 min	rt	82

[a] 53% of **1a** was recovered.
[b] 49% of **1a** was recovered.

Table 2. *t*-BuOK-catalyzed michael addition of 1,2-allenic ketones with 2-substituted diethyl malonates

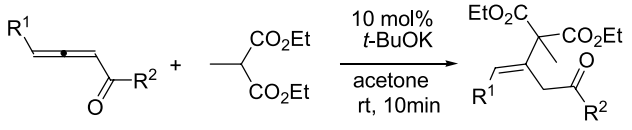


1 **4** **5**

Entry	1		4	Yield of **5** (%)
	R^3	R^4	R	
1	Bn	Me (**1a**)	Allyl(**4b**)	85 (**5ab**)
2	Bn	Me (**1a**)	Bn (**4c**)	81 (**5ac**)
3	allyl	Me (**1b**)	Me (**4a**)	83 (**5ba**)
4	allyl	Me (**1b**)	Allyl (**4b**)	78 (**5bb**)
5	allyl	Me (**1b**)	Bn (**4c**)	76 (**5bc**)
6	*n*-C_4H_9	Me (**1c**)	Me (**4a**)	Trace
7[a]	*n*-C_4H_9	Me (**1c**)	Me (**4a**)	83 (**5ca**)
8	Ph	Bn (**1d**)	Me (**4a**)	64 (**5da**)
9	H	*n*-C_8H_{17} (**1e**)	Me (**4a**)	66 (**5ea**)
10	H	*n*-C_7H_{15} (**1f**)	Me (**4a**)	62 (**5fa**)
11	H	H (**1 g**)	Me (**4a**)	44 (**5ga**)

[a] 100 mol% of *t*-BuOK was used.

afforded **5ca** in 83% yield (Table 2, entries 6–7). Similarly, **1e**–**1f** also reacted with 2-methyl diethyl malonate in the presence of 10 mol% *t*-BuOK to afford **5ea** and **5fa** in moderate yield (Table 2, entries 9–10). The Michael addition can be further applied to γ-substituted allenic ketones. Under similar reaction conditions, the simplest 1,2-allenic ketone **1g** reacted with 2-methyl diethyl malonate **4a** affording **5ga** in a relatively low yield (44%) (Table 2, entry 11). Reaction of **1h**–**1j** with **4a** also afforded β,γ-unsaturated enones *E*-**5ha**, *E*-**5ia**, and *E*-**5ja** as sole product (Scheme 4). The stereochemistry of this product was determined by the ^1H–^1H NOESY spectra (Fig. 1).



1 **4a** **5**

R^1	R^2	yield of (*E*)-**5** (%)
n-C_4H_9	CH_3 (**1h**)	49 ((*E*)-**5ha**)
n-C_7H_{15}	CH_3 (**1i**)	59 ((*E*)-**5ia**)
n-C_7H_{15}	*n*-C_7H_{15} (**1j**)	53 ((*E*)-**5ja**)

Scheme 4.

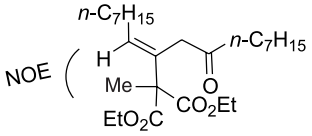
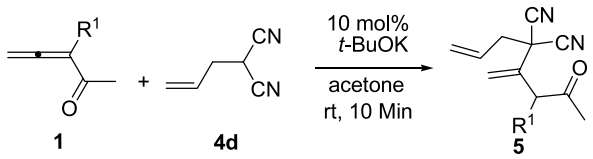


Figure 1. ^1H–^1H NOE effect of product *E*-**5ja**.

In addition, this transformation may be extended to other 2-substituted active methylene compounds. For example, 2-substituted malononitrile is also an effective nucleophile. In this case, α-substituted allenic ketones **1a**, **1b** and **1c** reacted with 2-allyl malononitrile **4d** to afford the corresponding products **5ad**, **5bd** and **5cd** in relative good yields (Scheme 5). However, when 2-allyl substituted



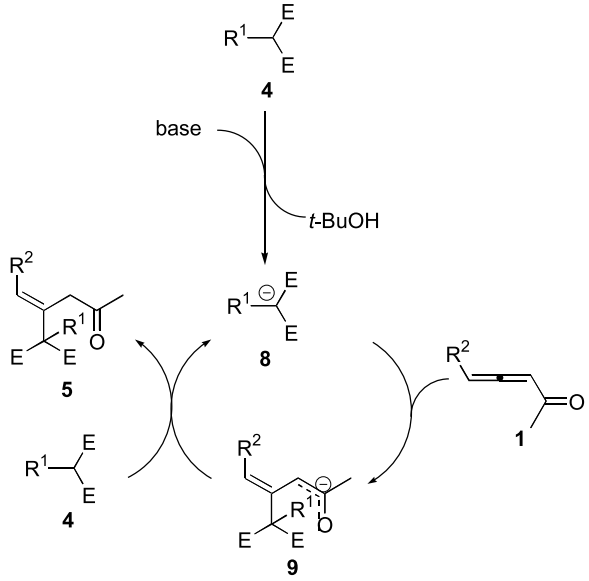
1 **4d** **5**

R^1	yield of **7**(%)
B*n* (**1a**)	65 (**5ad**)
Allyl (**1b**)	63 (**5bd**)
n-C_4H_9 (**1c**)	79 (**5cd**)

Scheme 5.

acetoacetone, nitroacetic acid ethyl ester and benzenesulfonylacetic acid ethyl ester were treated similarly, complicated reactions were observed.

A rationale for the catalytic reaction is shown in Scheme 6.



Scheme 6.

t-BuOK plays a role as an initiator to generate anion **8**, which adds to the conjugated carbon–carbon double bond in 1,2-allenic ketones from the less hindered side to form intermediate **9**. Immediate proton transfer between **9** and **4**, which may prevent the formation of α,β-unsaturated enones, affords products **5** and regenerates malonate anion **8** (Scheme 6).

In summary, we have demonstrated a facile nucleophilic addition of 2-substituted diethyl malonates or malononitriles to various substituted 1,2-allenic ketones to afford β,γ-unsaturated enones using *t*-BuOK as the basic catalyst. The reaction usually completes in 10 min in acetone at room temperature. With the γ-substituted allenic ketones the *E*-isomer was afforded highly stereoselectively.

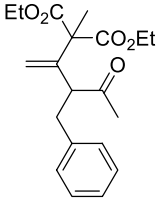
2. Experimental

^{1}H and ^{13}C NMR spectra were recorded on a Mercury 300 spectrometer using $CDCl_3$ as the solvent and TMS as the internal standard. Mass spectra were obtained using HP 5989A spectrometers. IR spectra were measured on Avatar 360 spectrophotometer. High-resolution mass spectra were carried out with Concept 1H spectrometer. TLC was performed on pre-coated plates (0.25 mm, silica gel 60 F_{254}).

2.1. General procedure for *t*-BuOK-catalyzed 1,4-addition of diethyl 2-substituted malonate with 1,2-allenic ketones

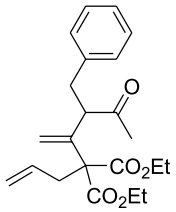
A solution of **1** (1.0 mmol), diethyl 2-substitutedmalonate or malononitrile (1.0–1.2 mmol), and 11 mg (10 mol%) of *t*-BuOK in 2 mL of acetone was stirred at room temperature. The reaction usually completed within 10 min as monitored by TLC. After the reaction was over, the solvent was evaporated and the crude product was purified by chromatography on silical gel (petroleum ether /ethyl acetate 10/1) to afford **5**.

2.1.1. 3-Benzyl-4-(1′,1′-bis(ethoxycarbonyl)ethyl)-4-penten-2-one (5aa).



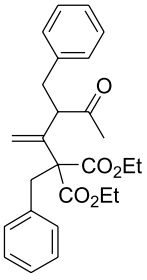
The reaction of **1a** (202 mg, 1.2 mmol), *t*-BuOK (12 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 282 mg (82%) of **5aa**: yellow liquid; IR (neat) 1731, 1637 cm^{-1}; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.29–7.14(m, 5H), 5.51 (s, 1H), 5.34 (s, 1H), 4.21 (q, J=7.2 Hz, 4H), 3.67 (dd, J_1=10.2 Hz, J_2=4.2 Hz, 1H), 3.06 (dd, J_1=13.5 Hz, J_2=10.2 Hz, 1H), 2.88 (dd, J_1=13.5 Hz, J_2=4.2 Hz, 1H), 1.96 (s, 3H), 1.62 (s, 3H), 1.28–1.22 (m, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 208.3, 170.9, 170.7, 143.5, 139.5, 128.8, 128.4, 126.3 117.0, 61.73, 61.66, 60.3, 57.0, 41.0, 30.6, 20.8, 13.93, 13.90; MS *m*/*z* (%) 43 (100), 346 (M^+, 2.92); HRMS *m*/*z* (MALDI) calcd for $C_{20}H_{26}O_5Na^+$(M^++Na) 369.1678. Found 369.1673.

2.1.2. 3-Benzyl-4-methylene-5,5-bis(ethoxycarbonyl)-7-octen-2-one (5ab).



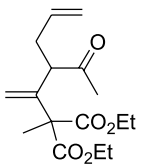
The reaction of **1a** (172 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-allylmalonate **4b** (200 mg, 1.0 mmol) in 2 mL of acetone afforded 316 mg (85%) of **5ab**: yellow liquid; IR (neat) 1731, 1638 cm^{-1}; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.26–7.09 (m, 5H), 5.84–5.68 (m, 1H), 5.56 (s, 1H), 5.46 (s, 1H), 5.12–4.97 (m, 2H), 4.24–4.03 (m, 4H), 3.69 (dd, J_1=10.2 Hz, J_2=4.5 Hz, 1H), 2.98 (dd, J_1=10.2 Hz, J_2=13.2 Hz, 1H), 2.89 (d, J=4.5 Hz, 1H), 2.85 (d, J=6.6 Hz, 2H), 1.87 (s, 3H), 1.22 (m, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.8, 169.5, 169.3, 141.7, 139.3, 132.9, 128.7, 128.2, 126.2, 118.5, 118.3, 63.7, 61.5, 61.4, 57.0, 41.0, 38.4, 30.5, 13.84, 13.80; MS *m*/*z* (%) 91 (100), 331 ($M^+-C_3H_5$, 9.89); HRMS *m*/*z* (MALDI) calcd for $C_{22}H_{28}O_5Na^+$(M^++Na) 395.1834. Found 395.1829.

2.1.3. 3-Benzyl-4-(1′,1′-bis(ethoxycarbonyl)-2′-phenylethyl)-4-penten-2-one (5ac).



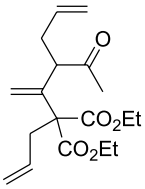
The reaction of **1a** (172 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-benzylmalonate **4c** (250 mg, 1.0 mmol) in 2 mL of acetone afforded 342 mg (81%) of **5ac**: yellow liquid; IR (neat) 1721, 1633 cm^{-1}; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.28–7.14 (m, 10H), 5.62 (s, 1H), 5.57 (s, 1H), 4.18–3.90 (m, 4H), 3.77 (dd, J_1=10.2 Hz, J_2=4.2 Hz, 1H), 3.50 (d, J=25.2 Hz, 1H), 3.45 (d, J=25.2 Hz, 1H), 3.03 (dd, J_1=10.2 Hz, J_2=13.2 Hz, 1H), 2.87 (dd, J_1=13.2 Hz, J_2=4.2 Hz, 1H), 1.90 (s, 3H), 1.18 (t, J=7.05 Hz, 3H), 1.10 (t, J=7.05 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.7, 169.4, 169.2, 142.8, 139.3, 136.2, 130.0, 128.8, 128.2, 127.8, 126.7, 126.2, 118.7, 64.9, 61.4, 61.3, 57.2, 40.61, 40.58, 30.3, 13.7, 13.5; MS *m*/*z* (%) 91 (100), 331 ($M^+-C_7H_7$, 48); HRMS *m*/*z* (MALDI) calcd for $C_{26}H_{30}O_5Na^+$(M^++Na) 445.1991. Found 445.1986.

2.1.4. 3-(1′-Methylene-2′,2′-bis(ethoxycarbonyl)propyl)-5-hexen-2-one (5ba).



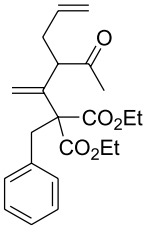
The reaction of **1b** (122 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 309 mg (83%) of **5ba**: yellow liquid; IR (neat) 1733, 1640 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.71–5.50 (m, 1H), 5.26 (s, 1H), 5.24 (s, 1H), 5.00–4.79 (m, 2H), 4.21–4.05 (m, 4H), 3.34 (dd, $J_1=9.3$ Hz, $J_2=5.4$ Hz, 1H), 2.57–2.40 (m, 1H), 2.28–2.09 (m, 1H), 1.93 (s, 3H), 1.55 (s, 3H), 1.16–1.26 (m, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.7, 170.8, 170.5, 143.3, 135.5, 116.7, 116.5, 61.54, 61.50, 60.1, 55.2, 38.3, 29.2, 20.7, 13.78, 13.76; MS (CI) *m/z* (%) 181 (100), 297 (M^+ + H, 9); HRMS *m/z* (MALDI) calcd for $C_{16}H_{24}O_5Na^+$(M^+ + Na) 319.1521. Found 319.1516.

2.1.5. 3-Allyl-4-methylene-5-bis(ethoxycarbonyl)-7-octen-2-one (5bb).



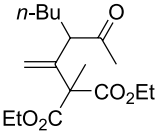
The reaction of **1b** (122 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-allylmalonate **4b** (200 mg, 1.0 mmol) in 2 mL of acetone afforded 251 mg (78%) of **5bb**: yellow liquid; IR (neat) 1728, 1640 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.82–5.62 (m, 2H), 5.44 (s, 1H), 5.39 (s, 1H), 5.12–4.85 (m, 4H), 4.25–4.08 (m, 4H), 3.41 (dd, $J_1=9.3$ Hz, $J_2=5.1$ Hz, 1H), 2.84 (d, $J=7.2$ Hz, 2H), 2.57–2.43 (m, 1H), 2.34–2.21 (m, 1H), 2.17 (s, 3H), 1.24 (m, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.4, 169.5, 169.2, 141.7, 135.5, 132.9, 118.5, 118.2, 116.6, 63.7, 61.43, 61.38, 55.4, 38.5, 38.4, 29.3, 13.8; MS (CI) *m/z* (%) 235 (100), 323 (M^++H, 19); HRMS *m/z* (MALDI) calcd for $C_{18}H_{26}O_5Na^+$(M^++Na) 345.1678. Found 345.1673.

2.1.6. 3-(1′-Methylene-2′,2′-bis(ethoxycarbonyl)-3′-phenylpropyl)-5-hexen-2-one (5bc).



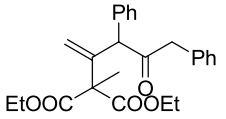
The reaction of **1b** (122 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-benzylmalonate **4c** (250 mg, 1.0 mmol) in 2 mL of acetone afforded 283 mg (76%) of **5bc**: yellow liquid; IR (neat) 1732, 1640 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 7.16 (br, 5H), 5.67–5.55 (m, 1H), 5.50 (s, 1H), 5.38 (s, 1H), 5.00–4.90 (m, 2H), 4.15–3.79 (m, 4H), 3.50–3.32 (m, 3H), 2.51–2.39 (m, 1H), 2.27–2.16 (m, 1H), 2.10 (s, 3H), 1.14 (t, $J=6.9$ Hz, 3H), 1.05 (t, $J=7.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.2, 169.3, 169.1, 142.7, 136.2, 135.4, 130.0, 127.8, 126.7, 118.5, 116.5, 64.9, 61.32, 61.27, 55.6, 40.5, 38.2, 29.2, 13.6, 13.5; MS *m/z* (%) 91 (100), 331 ($M^+-C_3H_5$, 31); HRMS *m/z* (MALDI) calcd for $C_{22}H_{28}O_5Na^+$(M^+ + Na) 395.1834. Found 395.1829.

2.1.7. 3-Butyl-4-(1′,1′-bis(ethoxycarbonyl)ethyl)-4-penten-2-one (5ca).



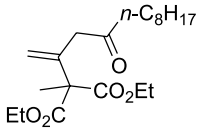
The reaction of **1c** (138 mg, 1.0 mmol), *t*-BuOK (112 mg, 1.0 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 259 mg (83%) of **5ca**: yellow liquid; IR (neat) 1732, 1637 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.31 (s, 1H), 5.28 (s, 1H), 4.27–4.10 (m, 4H), 3.24 (dd, $J_1=8.4$ Hz, $J_2=6.0$ Hz, 1H), 2.19 (s, 3H), 1.84–1.70 (m, 1H), 1.60 (s, 3H), 1.59–1.44 (m, 1H), 1.31–1.14 (m, 10H), 0.85 (t, $J=7.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 208.2, 170.5, 170.3, 144.0, 115.7, 61.21, 61.19, 60.0, 55.7, 45.7, 33.7, 29.9, 28.3, 22.4, 20.6, 13.53, 13.52; MS *m/z* (%) 239 (100), 312 (M^+, 0.82); HRMS *m/z* (EI) calcd for $C_{17}H_{28}O_5$ 312.1937. Found 312.1966.

2.1.8. 1, 3-Diphenyl-4-(1′,1′-bis(ethoxycarbonyl)ethyl)-4-penten-2-one (5da).



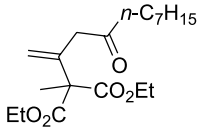
The reaction of **1d** (234 mg, 1 mmol), *t*-BuOK (11.0 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (209 mg, 1.2 mmol) in 2 mL of acetone afforded 261 mg (64%) of **5da**: yellow liquid; IR (neat) 1728, 1633 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 7.34–7.02 (m, 10H), 5.50 (s, 1H), 5.29 (s, 1H), 5.04 (s, 1H), 4.20 (q, $J=7.2$ Hz, 2H), 3.95 (s, 2H), 3.92–3.77 (m, 2H), 1.57 (s, 3H), 1.28 (t, $J=7.2$ Hz, 3H), 1.12 (t, $J=7.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 204.7, 171.1, 170.7, 141.9, 137.0, 134.4, 129.7, 128.9, 128.3, 127.1, 126.7, 119.2, 61.7, 61.5, 60.3, 59.8, 48.8, 20.7, 13.9, 13.7; MS *m/z* (%) 91 (100), 408(M^+, 0.72); HRMS *m/z* (MALDI) calcd for $C_{25}H_{28}O_5Na^+$(M^+ + Na) 431.1834. Found 431.1829.

2.1.9. 2-(1′,1′-Bis(ethoxycarbonyl)ethyl)-1-dodecen-4-one (5ea).



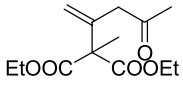
The reaction of **1e** (180 mg, 1.0 mmol), *t*-BuOK (11.0 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 234 mg (66%) of **5ea**: yellow liquid; IR (neat) 1732, 1639 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.29 (s, 1H), 5.13 (s, 1H), 4.23–4.12 (m, 4H), 3.29 (s, 2H), 2.47 (t, $J=7.5$ Hz, 2H), 1.61 (s, 3H), 1.60–1.49 (m, 2H), 1.29–1.18 (m, 16H), 0.87 (t, $J=6.5$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.9, 170.7, 139.1, 118.0, 61.6, 59.5, 47.8, 42.0, 31.6, 29.1, 29.0, 23.7, 22.5, 20.4, 14.0, 13.9; MS *m/z* (%) 181 (100), 354 (M^+, 0.82); HRMS *m/z* (EI) calcd for $C_{20}H_{34}O_5(M^+)$ 354.2406. Found 354.2438.

2.1.10. 2-(1′,1′-Bis(ethoxycarbonyl)ethyl)-1-undecen-4-one (5fa).



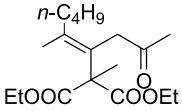
The reaction of **1f** (166 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 211 mg (62%) of **5fa**: yellow liquid; IR (neat) 1732, 1639 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.30 (s, 1H), 5.13 (s, 1H), 4.24–4.12 (m, 4H), 3.30 (s, 2H), 2.48 (t, $J=6.9$ Hz, 2H), 1.62 (s, 3H), 1.60–1.49 (m, 2H), 1.38–1.20 (m, 14H), 0.88 (t, $J=6.0$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 207.8, 170.6, 139.0, 117.9, 61.5, 59.4, 47.7, 41.9, 31.7, 29.2, 29.01, 28.99, 23.6, 22.5, 20.3, 13.9, 13.8; MS *m/z* (%) 57 (100), 340 (M^+, 0.61); HRMS *m/z* (EI) calcd for $C_{19}H_{32}O_5(M^+)$ 340.2244. Found 340.2219.

2.1.11. 4-(1′,1′-Bis(ethoxycarbonyl)ethyl)-4-penten-2-one (5ga).



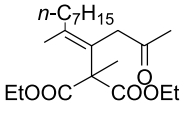
The reaction of **1g** (82 mg, 1.0 mmol), *t*-BuOK (11.0 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (209 mg, 1.2 mmol) in 2 mL of acetone afforded 113 mg (44%) of **5ga**: yellow liquid; IR (neat) 1732, 1639 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.31 (s, 1H), 5.15 (s, 1H), 4.18 (q, $J=$ 6.6 Hz, 2H), 4.19 (q, $J=6.6$ Hz, 2H), 3.31 (s, 2H), 2.19 (s, 3H) 1.62 (s, 3H), 1,26 (t, $J=6.6$ Hz, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 205.7, 170.5, 139.0, 118.2, 61.5, 59.3, 48.7, 29.0, 20.2, 13.7; MS *m/z* (%) 43 (100), 257(M^+ +H, 0.81); HRMS *m/z* (EI) calcd for $C_{13}H_{20}O_5(M^+)$ 256.1344. Found 256.1336.

2.1.12. 4-(1′, 1′-Bis(ethoxycarbonyl)ethyl)-4-nonen-2-one ((*E*)-5ha).



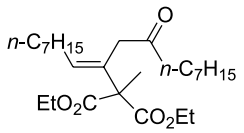
The reaction of **1h** (138 mg, 1 mmol), *t*-BuOK (12.0 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (209 mg, 1.2 mmol) in 2 mL of acetone afforded 153 mg (49%) of (*E*)-**5ha**: yellow liquid; IR (neat) 1731, 1638 cm^{-1}; ^1H NMR (300 MHz, CDCl3): δ 5.64 (t, $J=6.9$ Hz, 1H), 4.15 (q, $J=7.2$ Hz, 2H), 4.14 (q, $J=7.2$ Hz, 2H), 3.36 (s, 2H), 2.15 (s, 3H), 1.95 (q, $J=7.2$ Hz, 2H), 1.58 (s, 3H), 1.42–1.22 (m, 4H), 1.23 (t, $J=7.2$ Hz, 6H), 0.87 (t, $J=6.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 204.9, 171.2, 133.0, 129.1, 61.3, 59.8, 44.9, 31.0, 29.2, 28.5, 22.2, 20.3, 13.81, 13.78; MS *m/z* (%) 43 (100), 312 (M^+, 0.84); HRMS *m/z* (EI) calcd for $C_{17}H_{28}O_5(M^+)$ 312.1937. Found 312.1918.

2.1.13. 4-(1′,1′-Bis(ethoxycarbonyl)ethyl)-4-dodecen-2-one ((*E*)-5ia).



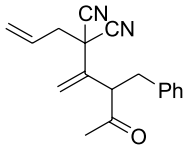
The reaction of **1i** (180 mg, 1 mmol), *t*-BuOK (11.0 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 207 mg (59%) of (*E*)-**5ia**: yellow liquid; IR (neat) 1732, 1669 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.64 (t, $J=6.9$ Hz, 1H), 4.15 (q, $J=7.2$ Hz, 2H), 4.14 (q, $J=7.2$ Hz, 2H), 3.36 (s, 2H), 2.14 (s, 3H), 1.93 (q, $J=7.2$ Hz, 2H), 1.58 (s, 3H), 1.39–1.29 (m, 2H), 1.26–1.19 (m, 14H), 0.86 (t, $J=6.3$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 205.0, 171.2, 133.1, 129.0, 61.4, 59.8, 44.9, 31.7, 29.3, 29.2, 29.1, 28.9, 28.8, 26.8, 22.56, 22.54, 20.3, 14.0, 13.8; MS *m/z* (%) 43 (100), 354 (M^+, 0.31); HRMS *m/z* (MALDI) calcd for $C_{20}H_{34}O_5Na^+$ (M^+ +Na) 377.2304. Found 377.2299.

2.1.14. 10-(1′,1′-Bis(ethoxycarbonyl)ethyl)-10-octadecen-8-one ((*E*)-5ja).



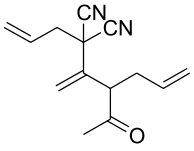
The reaction of **1j** (264 mg, 1.0 mmol), *t*-BuOK (11 mg, 0.1 mmol) and diethyl 2-methylmalonate **4a** (174 mg, 1.0 mmol) in 2 mL of acetone afforded 232 mg (53%) of *E*-**5ja**: yellow liquid; IR (neat) 1732 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.63 (t, $J=6.9$ Hz, 1H), 4.15 (q, $J=$ 7.2 Hz, 2H), 4.14 (q, $J=7.2$ Hz, 2H), 3.35 (s, 2H), 2.41 (t, $J=7.5$ Hz, 2H), 1.92 (q, $J=6.9$ Hz, 2H), 1.58 (s, 3H), 1.57–1.50 (m, 2H), 1.42–1.18 (m, 24H), 0.87 (t, $J=6.3$ Hz, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 206.9, 171.3, 132.9, 129.0, 61.3, 59.9, 44.1, 42.1, 31.7, 31.6, 29.2, 29.11, 29.05, 29.0, 28.92, 28.88, 23.8, 22.54, 22.51, 20.3, 13.99, 13.97, 13.8; MS *m/z* (%) 265 (100), 438 (M^+, 0.72); HRMS *m/z* (EI) calcd for $C_{26}H_{46}O_5$ 438.3345. Found 438.3343.

2.1.15. 3-Benzyl-4-methylene-5,5-biscyano-7-octen-2-one (5ad).



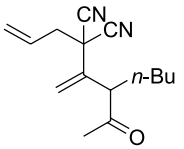
The reaction of **1a** (172 mg, 1 mmol), *t*-BuOK (11.0 mg, 0.1 mmol) and 2-allyl malononitrile (106 mg, 1.0 mmol) in 2 mL of acetone afforded 180 mg (65%) of **5ad**: yellow liquid; IR (neat) 3087, 1720, 1643 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 7.33–7.14 (m, 5H), 5.82 (s, 1H), 5.81–5.65 (m, 1H), 5.66 (s, 1H), 5.40 (d, $J=14.1$ Hz, 1H), 5.36 (d, $J=20.4$ Hz, 1H), 3.64 (dd, $J_1=5.7$ Hz, $J_2=9.3$ Hz, 1H), 3.24 (dd, $J_1=9.3$ Hz, $J_2=13.8$ Hz, 1H), 2.90 (dd, $J_1=5.7$ Hz, $J_2=13.8$ Hz, 1H), 2.57 (d, $J=6.9$ Hz, 2H), 2.12 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 204.8, 137.9, 136.0, 128.9, 128.7, 127.9, 127.0, 123.6, 120.5, 113.8, 113.5, 56.1, 44.1, 41.5, 39.7, 30.0; MS *m/z* (%) 43 (100), 279 (M^+, 1.67); HRMS *m/z* (MALDI) calcd for $C_{18}H_{19}N_2O^+$ (M^+ + H) 279.1497. Found 279.1492.

2.1.16. 3-Allyl-4-methylene-5,5-biscyano-7-octen-2-one (5bd).



The reaction of **1b** (122 mg, 1 mmol), *t*-BuOK (12.0 mg, 0.1 mmol) and 2-allyl malononitrile (106 mg, 1.0 mmol) in 2 mL of acetone afforded 143 mg (63%) of **5bd**: yellow liquid; IR (neat) 3084, 1720, 1643 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.76 (s, 1H), 5.92–5.58 (m, 2H), 5.45 (s, 1H), 5.42–5.36 (m, 2H), 5.09–5.03 (m, 2H), 3.36 (t, $J=6.9$ Hz, 1H), 2.73 (d, $J=6.6$ Hz, 2H), 2.69–2.57 (m, 1H), 2.37–2.25 (m, 1H), 2.22 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 204.5, 135.8, 133.8, 127.9, 123.6, 120.3, 118.2, 113.7, 113.4, 54.0, 44.0, 41.6, 37.2, 29.1; MS *m/z* (%) 43 (100), 228 (M^+, 1.84); HRMS *m/z* (MALDI) calcd for $C_{14}H_{17}N_2O^+$(M^+ + H) 229.1341. Found 229.1335.

2.1.17. 3-Butyl-4-methylene-5,5-biscyano-7-octen-2-one (5cd).



The reaction of **1c** (138 mg, 1 mmol), *t*-BuOK (12.0 mg, 0.1 mmol) and 2-allyl malononitrile (106 mg, 1.0 mmol) in 2 mL of acetone afforded 193 mg (79%) of **5cd**: yellow liquid; IR (neat) 3088, 1720, 1644 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ 5.92–5.77 (m, 1H), 5.76 (s, 1H), 5.49 (s, 1H), 5.46–5.36 (m, 2H), 3.28 (t, $J=6.9$ Hz, 1H), 2.39 (d, $J=7.5$ Hz, 2H), 2.26 (s, 3H), 1.99–1.86 (m, 1H), 1.63–1.49 (m, 1H), 1.33–1.23 (m, 4H), 0.87 (t, $J=6.75$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ 205.4, 136.6, 128.0, 123.6, 119.6, 113.8, 113.6, 54.6, 44.2, 41.6, 33.2, 29.9, 28.9, 22.4, 13.6; MS *m/z* (%) 43 (100), 245 (M^++H, 3.97); HRMS *m/z* (MALDI) calcd for $C_{15}H_{21}N_2O^+$ (M^++H) 245.1654. Found 245.1648.

Acknowledgements

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